

After Simon, after Hrushovski

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ABSTRACT.

1. Patterns

In the following $\mathcal{U}$ is some monster model of a complete theory T of language L . For simplicity T is assumed to be one-sorted.

We define a new structure $\mathcal{N}$ whose domain is a subset of $\mathcal{U}^x$, where x is some fixed tuple of variables. The language of $\mathcal{N}$ is denoted by L^p . Though L^p depends on x , this will be left implicit. We also use an auxiliary tuple $z = \langle z_i : i < \lambda \rangle$ of (so-called) parameter variables.

Fix a set of formulas $L_z \subseteq \Delta(x; z) \subseteq L_{x,z}$. Let V be some large set of variables that are copies of x . Finally, let Δ be the set of Boolean combinations of formulas in $\bigcup_{x' \in V} \Delta(x'; z)$. The choice of Δ is justified in Remark 1.

Let $A \subseteq \mathcal{U}^x$ and $B \subseteq \mathcal{U}^z$. We write A, Δ for the set of formulas $\varphi(\bar{a}; z)$ for $\varphi(\bar{x}; z) \in \Delta$ and $\bar{a} \in A^{\bar{x}}$, and we write Δ, B for the set of formulas $\varphi(\bar{x}; b)$ for $\varphi(\bar{x}; z) \in \Delta$ and $b \in B^z$.

Let $\bar{a}, \bar{a}' \in (\mathcal{U}^x)^n$ and $b, b' \in \mathcal{U}^z$. If $\varphi(\bar{a}; b) \leftrightarrow \varphi(\bar{a}'; b)$ holds for every $\varphi(\bar{x}; z) \in \Delta$ and $b \in B^z$, we write $\bar{a} \equiv_{\Delta, B} \bar{a}'$. We define $b \equiv_{A, \Delta} b'$ similarly.

For every formula $\varphi(x_0, \dots, x_{n-1}; z) \in \Delta$ the language L^p contains the n -ary relation r_φ . Below by the symbol L^p we will also denote the set of *atomic* formulas $r_\varphi(x_0, \dots, x_{n-1})$ for any $x_0, \dots, x_{n-1} \in V$. Note that equalities are *not* included in L^p .

Given pairs of models $M \leq N$, where N is $|M|^+$ -saturated, we define a model $\mathcal{N}$ with domain N^x . We define the interpretation of r_φ in $\mathcal{N}$. If $\bar{a} = a_0, \dots, a_{n-1} \in \mathcal{N}$, we declare $r_\varphi(\bar{a})$ true if $\varphi(\bar{a}; b)$ holds for every $b \in M^z$ such that $\alpha(b)$. We repeat the definition for emphasis

$$r_\varphi(\bar{a}) \Leftrightarrow \varphi(\bar{a}; M^z) = M^z.$$

The following is paramount.

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1 Remark By the choice of Δ , the truth of $r_\varphi(\bar{a})$ only depends on the equivalence class of $\equiv_{\Delta, M}$ of the *components* of the tuple $\bar{a}$. Explicitly, if $\bar{a} = a_0, \dots, a_{n-1}$ and $\bar{a}' = a'_0, \dots, a'_{n-1}$ are such that $a_i \equiv_{\Delta, M} a'_i$ for every $i < n$, then

$$\# \quad r_\varphi(\bar{a}) \leftrightarrow r_\varphi(\bar{a}').$$

By the remark, it would be natural to define the interpretation of L^P on the quotient $\mathcal{N}/\equiv_{\Delta, M}$. But we prefer to keep the original domain $\mathcal{N}$ to have a common playground with L .

A **p-type** is a set of (atomic) formulas in $L^P(\mathcal{N})$. We write **p-tp**($\bar{a}$) for the p-type of $\bar{a}$, i.e. the set of formulas $r_\varphi(\bar{x})$ such that $r_\varphi(\bar{a})$ holds.

Note incidentally that the relation $x_0 \equiv_{\Delta, M} x_1$ is definable by a p-type. Namely the p-type containing the formulas $r_\vartheta(x_0, x_1)$ for every $\vartheta(x_0, x_1; z)$ of the form $\varphi(x_0; z) \leftrightarrow \varphi(x_1; z)$.

2 Fact Formulas in L^P are closed under adjunction of dummy variables, disjunction, conjunction, and existential quantification.

Proof. The claim about dummy variables is clear, so we tacitly use it below.

We say that $\varphi'(\bar{x}; z)$ is an alphabetic variant of $\varphi(\bar{x}; z)$ if it is obtained by replacing the variables effectively occurring in $\varphi(\bar{x}; z)$ by other variables among those in z . Clearly, if $\varphi'(\bar{x}; z)$ is an alphabetic variant of $\varphi(\bar{x}; z)$, then $r_\varphi(\bar{x})$ and $r_{\varphi'}(\bar{x})$ are equivalent.

Up to replacing $\varphi(\bar{x}; z)$ with an alphabetic variant, we can assume that the z -variables are disjoint from those in $\psi(\bar{x}; z)$. Therefore the following equivalences hold

$$\begin{aligned} r_\varphi(\bar{a}) \vee r_\psi(\bar{a}) &\Leftrightarrow \forall z \in M^Z \varphi(\bar{a}; z) \vee \forall z \in M^Z \psi(\bar{a}; z) \\ &\Leftrightarrow \forall z \in M^Z [\varphi(\bar{a}; z) \vee \psi(\bar{a}; z)] \\ &\Leftrightarrow r_{\varphi \vee \psi}(\bar{a}) \end{aligned}$$

A similar argument applies to the conjunction of two formulas.

Finally, consider the formula $\exists y r_\varphi(\bar{x}, y)$. Without loss of generality, we can assume that φ has the form

$$\varphi(\bar{x}, y; z) = \bigvee_{i=1}^m \varphi_i(\bar{x}, y; z)$$

where $\varphi_i(\bar{x}, y; z) = \varphi'_i(\bar{x}; z) \wedge \psi_i(y; z)$ and each $\varphi'_i(\bar{x}; z)$ is a conjunction of formulas in $\bigcup_{x' \in V} L_{x'; z}$ or negation thereof. Closure under existential quantification follows from the following equivalences

$$\begin{aligned} \exists y r_\varphi(\bar{a}, y) &\Leftrightarrow \exists y \forall z \in M^Z \bigvee_{i=1}^m \varphi_i(\bar{a}, y; z) \\ &\Leftrightarrow \forall z \in M^Z \bigvee_{i=1}^m [\varphi'_i(\bar{a}; z) \wedge \exists y \psi_i(y; z)] \end{aligned}$$

$$\Leftrightarrow r_{\vartheta}(\bar{a}),$$

where

$$\vartheta(\bar{x}; z) = \bigwedge_{i=1}^m \varphi'_i(\bar{x}; z) \wedge \exists y \psi_i(y; z)$$

is a Boolean combination of formulas in $\bigcup_{x' \in V} L_{x'; z}$, as required. $\square$

3 Fact (∞ -p-saturation) Let $A \subseteq \mathcal{N}$ have arbitrary cardinality. Let $\bar{x}$ be a tuple of copies of x of length $\leq |\mathcal{N}|$. Then every p-type $p(\bar{x}) \subseteq L^p(A)$ that is finitely consistent in $\mathcal{N}$ is realized in $\mathcal{N}$.

Proof. Let $A' \subseteq \mathcal{N}$ be a small set containing a representative of every $\equiv_{\Delta, M}$ equivalence class that intersects A . Replacing the elements of A that occur in $p(\bar{x})$ by elements of A' we obtain an equivalent type $p'(\bar{x}) \subseteq L^p(A')$. Note that every formula $r_{\varphi}(\bar{x})$ there is a Δ -type $q(\bar{x}) \subseteq L(M)$ such that

$$\mathcal{N} \models r_{\varphi}(\bar{a}) \Leftrightarrow \mathcal{N} \models q(\bar{a})$$

Therefore the saturation of $\mathcal{N}$ implies that $p'(\bar{x})$ is realized in $\mathcal{N}$. $\square$

2. Morphism

A p-morphism is a map $f : \mathcal{N} \rightarrow \mathcal{N}$ such that for every $r_{\varphi}(\bar{x}) \in L^p$ and every $\bar{a} \in (\text{dom } f)^{\bar{x}}$

$$\# \quad \mathcal{N} \models r_{\varphi}(\bar{a}) \Rightarrow \mathcal{N} \models r_{\varphi}(f\bar{a}).$$

As $\equiv_{\Delta, M}$ is definable by a p-type, p-morphisms induce well-defined maps on the quotient $\mathcal{N}/\equiv_{\Delta, M}$. We say that two p-morphisms f and g are **equivalent**, and write $f \sim g$, if they induce the same map on $\mathcal{N}/\equiv_{\Delta, M}$. We often confuse equivalent p-morphisms and we use the same symbol f to denote the map induced on the quotient.

A total p-morphism is called a p-endomorphism. We write $\text{End}(\mathcal{N})$ for the set of p-endomorphisms of $\mathcal{N}$. We endow $\text{End}(\mathcal{N})$ with the topology that has as basic closed sets those of the form

$$\left\{ f \in \text{End}(\mathcal{N}) : \mathcal{N} \models r_{\varphi}(fa_1, \dots, fa_n, c_1, \dots, c_m) \right\}$$

for $\varphi(x; z) \in \Delta$ and $a_1, \dots, a_n, c_1, \dots, c_m \in \mathcal{N}$. Note that, by Remark 1, if $f \sim g$ then f and g belong to the same basic closed sets, therefore this topology naturally induces a topology on $\text{End}(\mathcal{N})/\sim$.

4 Fact With the above topology $\text{End}(\mathcal{N})$ is T_1 (singletons are closed) and compact. Composition of p-morphisms is a left continuous operation.

Proof. We prove that singletons are closed. In fact

$$\{f\} = \bigcap_{a \in \mathcal{N}} \left\{ g \in \text{End}(\mathcal{N}) : a \equiv_{\Delta, M} ga \right\}$$

therefore, as $\equiv_{\Delta, M}$ is defined by a p-type, $\{f\}$ is the intersection of closed sets.

We prove compactness. Assume the following intersection is finitely consistent (we have slightly compressed the notation)

$$\#. \quad \bigcap_{i < \lambda} \left\{ f \in \text{End}(\mathcal{N}) : \mathcal{N} \models r_i(f\bar{a}, \bar{c}) \right\}.$$

then the p-type

$$\bar{x} \equiv_{\Delta, M} \bar{a} \cup \{r_i(\bar{x}, \bar{c}) : i < \lambda\}$$

is finitely consistent in $\mathcal{N}$. By ∞ -p-saturation, it is realized by some $\bar{a}'$. By Fact 5 there is a p-endomorphism $\mathcal{N}$ that maps $\bar{a}$ to $\bar{a}'$. This proves that the set in # is non empty.

We prove left continuity of composition. That is for any given $f \in \text{End}(\mathcal{N})$ we prove that the function $\tau_f : \text{End}(\mathcal{N}) \rightarrow \text{End}(\mathcal{N})$ that maps $g \mapsto g \circ f$ is continuous. Pick any basic closed set $\{h : r(h\bar{a}, \bar{c})\}$. Then

$$\tau_f^{-1}[\{h : r(h\bar{a}, \bar{c})\}] = \{g : r(g(f\bar{a}), \bar{c})\}$$

is closed, as required. $\square$

5 Fact Every p-morphism extends to a p-morphism defined on the whole of $\mathcal{N}$. Also, every p-isomorphism extends to a p-automorphism of $\mathcal{N}$.

Proof. Let $f : \mathcal{N} \rightarrow \mathcal{N}$ be a p-morphism. Let $\bar{a}$ be an enumeration of $\text{dom } f$ and let $b \in \mathcal{N}$. Let $p(\bar{x}) = \text{p-tp}(\bar{a})$ and let $q(\bar{x}, y) = \text{p-tp}(\bar{a}, b)$. By ∞ -p-saturation $\exists y q(\bar{x}, y)$ is equivalent to the type containing the existential closure of the formulas in $q(\bar{x}, y)$. By Fact 2, the latter is equivalent to a p-type, which is necessarily coincides with $p(\bar{x})$. Then $q(f\bar{a}, y)$ is consistent hence by ∞ -p-saturation it is realized by some $c \in \mathcal{N}$. Then $f \cup \{\langle b, c \rangle\}$ is a p-morphism defined in b that extends f .

The second claim is obtained by back-and-forth. $\square$

3. Duality

In this section $\hat{m}$ is an enumeration of M and $|z| = |\hat{m}|$. For a more convenient notation, x is taken of infinite length – then, up to equivalence, formulas in $L_z(\mathcal{U})$ have the form $\varphi(a; z)$ for some $\varphi(x; z) \in L$ and $a \in \mathcal{U}^x$.

6 Remark We stress a simple fact which is used in the proof below. Recall that $\varphi'(a; z)$ is an alphabetic variant of $\varphi(a; z)$ if it is obtained by replacing the variables effectively occurring in $\varphi(a; z)$ by other variables among those in z . The following are clearly equivalent

1. $\varphi(a; M^z) = M^z$
2. $\varphi'(a; \hat{m})$ holds for every alphabetic variant of $\varphi(a; z)$.

Let $\mathcal{B} = \{b \in \mathcal{U}^z : \hat{m} \equiv b \downarrow_M \mathcal{N}\}$, where $\equiv$ is the elementary equivalence in L however, as $L_z \subseteq \Delta$, we could replace it with $\equiv_{\emptyset, \Delta}$. The N, Δ -topology on $\mathcal{B}$ is the one generated

by the closed sets of the form $\{b \in \mathcal{B} : \varphi(\bar{a}; b)\}$ for $\varphi(\bar{x}; z) \in \Delta$ and $\bar{a} \in \mathcal{N}^{\bar{x}}$. The N, Δ -topology on $\mathcal{B}/\equiv_{N, \Delta}$ is the quotient of that on $\mathcal{B}$.

7 Fact The topological spaces $\text{End}(\mathcal{N})/\sim$ and $\mathcal{B}/\equiv_{N, \Delta}$ are homeomorphic.

Proof. In 1 below, we define a map $f \mapsto b_f$ from $\text{End}(\mathcal{N})$ into $\mathcal{B}$. In 2, we define a map $b \mapsto f_b$ from $\mathcal{B}$ into $\text{End}(\mathcal{N})$. The reader may check that these maps induce maps between $\text{End}(\mathcal{N})/\sim$ and $\mathcal{B}/\equiv_{N, \Delta}$ that are each other's inverse. In 3 we prove the continuity of the first map and in 4 the continuity of the second.

1. Let $f : \mathcal{N} \rightarrow \mathcal{N}$ be an p-epimorphism. Let $p(z)$ be the type containing the formulas $\varphi(a; z)$ for $\varphi(x; z) \in \Delta$ and $a \in \mathcal{N}$ such that $\varphi(fa; \hat{m})$. Clearly $p(z) \vdash \hat{m} \equiv z$ and is complete over N, Δ . We prove that $p(z)$ is finitely satisfied in M , then b_f is some/any realization of $p(z)$.

If for a contradiction that $p(z)$ is not finitely satisfied in M , then $\psi(a_1, \dots, a_n; M^z) = \emptyset$, where

$$\psi(x_1, \dots, x_n; z) = \bigwedge_{i=1}^n \varphi_i(x_i; z)$$

for some $\varphi_i(x; z) \in \Delta$ and $a_i \in \mathcal{N}$ such that $\varphi_i(fa_i; \hat{m})$ for every i . Then $r_{\neg\psi}(a_1, \dots, a_n)$ and $\neg r_{\neg\psi}(fa_1, \dots, fa_n)$. This is not possible because f is a p-morphism.

2. Let $b \in \mathcal{B}$ be given. We define

$$p(x, y, z) = \{\varphi(y; z) \rightarrow \varphi(x; \hat{m}) : \varphi(x; z) \in \Delta\}.$$

Note that $p(x, a, b)$ is a p-type, for any $a \in \mathcal{N}$ and $b \in \mathcal{B}$. We claim that if $a \equiv_{\Delta, M} a'$ then $p(x, a, b) = p(x, a', b)$. In fact, $\varphi(a; b) \leftrightarrow \varphi(a'; b)$ would contradict $b \perp_M N$. Also, note that $p(x, a, b)$ is consistent, otherwise $\neg \exists x \bigwedge_i \varphi_i(x; \hat{m})$, for some finitely many formulas $\varphi_i(x; z) \in \Delta$ such that $\varphi_i(a; b)$. But this would contradict $b \equiv \hat{m}$.

Then we can define $f_b(a)$ to be some/any realization of $p(x, a, b)$.

We prove that f_b is a p-morphism. Let $\psi(x_1, \dots, x_n; z) \in \Delta$ and $a_1, \dots, a_n \in \mathcal{U}^x$ be such that $r_\psi(a_1, \dots, a_n)$. Assume

$$\psi(x_1, \dots, x_n; z) = \bigvee_{h=1}^m \bigwedge_{k=1}^m \varphi_{h,k}(x_{i_{h,k}}; z)$$

for some $i_{h,k} \in \{1, \dots, n\}$. After re-indexing, from $r_\psi(a_1, \dots, a_n)$ we obtain $\varphi_k(a_k; M^z) = M^z$ for $k = 1, \dots, m$. Then from $b \perp_M N$ we obtain that b satisfies $\varphi_k(a_k; z)$ and all its alphabetic variants. Therefore $\hat{m}$ satisfies $\varphi_k(f_b a_k; z)$ and all its alphabetic variants. Therefore $\varphi_k(f_b a_k; M^z) = M^z$, and $r_\psi(f_b a_1, \dots, f_b a_n)$ follows.

3. We prove that the map defined in 1 is continuous. Pick any basic closed set in $\mathcal{B}$, say $\{b : \varphi(\bar{a}; b)\}$. We claim that the set $\{f : \varphi(\bar{a}; b_f)\}$ is closed in $\text{End}(\mathcal{N})$. Indeed, by the definition of b_f in 1

$$\{f : \varphi(\bar{a}; b_f)\} = \{f : \varphi(f\bar{a}; \hat{m})\}$$

$$\# = \{f : r_\psi(f\bar{a}, c)\}$$

where $\psi(\bar{x}, y; z)$ and c are chosen so that $r_\psi(\bar{x}, c)$ is equivalent to $\varphi(\bar{x}; \hat{m})$. As the set in $\#$ is closed in $\text{End}(\mathcal{N})$, continuity of the map $f \mapsto b_f$ follows.

We show how to choose $\psi(\bar{x}, y; z)$ and c . Let y be a copy of x . The formula $\psi(\bar{x}, y; z)$ is obtained from $\varphi(\bar{x}; z)$ by replacing the variables in the tuple z effectively occurring in $\varphi(\bar{x}; z)$ with variables in the tuple y . Note that z is dummy in $\psi(\bar{x}, y; z)$. Finally, we choose $c \in M^y$ such that $\psi(\bar{x}, c; z)$ is equivalent to $\varphi(\bar{x}; \hat{m})$.

4. We prove that the map defined in 2 is continuous. Pick any basic closed set in $\text{End}(\mathcal{N})$, say $\{h : r(h\bar{a}, \bar{c})\}$. We claim that the set $\{b : r(f_b\bar{a}, \bar{c})\}$ is closed. Indeed, by the definition of f_b in 2,

$$\{b : r(f_b\bar{a}, \bar{c})\} = \left\{b : b \models \exists \bar{x} \left[\bigwedge_{i=1}^n p(x_i, a_i, b) \wedge r(\bar{x}, \bar{c}) \right] \right\}$$

is closed in $\mathcal{B}$ because $r(\bar{x}, \bar{c})$ is definable by a N, Δ -type. $\square$

4. Semigroups

Assume $\mathcal{U}$ is a unital semigroup (a monoid) w.r.t. some definable operation which we denote by $\cdot$. The unit is denoted by 1. In this section we could work with $|x| = |z| = 1$ but, for uniformity with the previous section we take $|x| = |z| = \omega$. We write $z \cdot x$ for coordinatewise multiplication. In this section $\Delta(x, z)$ contains the formulas $\varphi(z \cdot x)$ for all $\varphi(x) \in L$.

8 Fact For $b \in \mathcal{B}$, let $g_b : \mathcal{N} \rightarrow \mathcal{N}$ be the function that maps $a \mapsto b \cdot a$. Then $b \mapsto g_b$ is an homeomorphism between $\mathcal{B} / \equiv_{\Delta, M}$ and $\text{End}(\mathcal{N}) / \sim$.

Proof. We prove that g_b is a p-morphism. Assume $r_{\psi, \alpha}(a_1, \dots, a_n)$. We prove $r_{\psi, \alpha}(g_b a_1, \dots, g_b a_n)$. Assume that for some $i_{h,k} \in \{1, \dots, n\}$

$$\psi(x_1, \dots, x_n; z) = \bigvee_{h=1}^m \bigwedge_{k=1}^m \varphi_{h,k}(z \cdot x_{i_{h,k}})$$

for some $\varphi_{h,k}(z \cdot x) \in \Delta$. From $r_\psi(a_1, \dots, a_n)$, after re-indexing, we obtain that $\varphi_k(M^z \cdot a_k) = M^z$ for $k = 1, \dots, m$. As $b \perp_M N$, we immediately obtain $\varphi_k(M^z \cdot b \cdot a_k) = M^z$ for all k , so $r_{\psi, \alpha}(g_b a_1, \dots, g_b a_n)$ follows. $\square$

5. Cores

A set $A \subseteq \mathcal{N}$ is **p-maximal** if it satisfies the following (obviously) equivalent conditions for every $\bar{a} \in A$ and every p-morphism f

- a. $r_\varphi(\bar{a}) \Leftrightarrow r_\varphi(f\bar{a})$ for every $r_\varphi \in L^p$
- b. $\text{p-tp}(\bar{a}) = \text{p-tp}(f\bar{a})$.

9 Fact The following are equivalent

1. $A \subseteq \mathcal{N}$ is p-maximal
2. $p(\bar{x}) = \text{p-tp}(\bar{a})$ is a maximally consistent p-type for every $\bar{a} \in A^{\bar{x}}$ and every finite $\bar{x}$
3. $p(\bar{y}) = \text{p-tp}(\bar{c})$ is a maximally consistent p-type for $\bar{c}$ any enumeration of A
4. every p-morphism defined on A is a p-isomorphism between A and its image.

Proof. $1 \Rightarrow 2$. Negate 2. Suppose $\text{p-tp}(\bar{a})$ is not maximal for some finite $\bar{a} \in A^{\bar{x}}$. Pick some $\bar{a}' \in A^{\bar{x}}$ such that $\text{p-tp}(\bar{a}) \subset \text{p-tp}(\bar{a}')$. Then $f = \{\langle \bar{a}, \bar{a}' \rangle\}$ is a p-morphism that contradicts the p-maximality of A .

$2 \Rightarrow 3$. If $\bar{c}$ is an enumeration of A that witnesses the failure of 3, then some finite subtuple of $\bar{c}$ witnesses the failure of 2.

$3 \Rightarrow 4$. Negate 4. Let f be a p-morphism defined on A that is not a p-isomorphism. Let $\bar{a} \in A^{\bar{x}}$ be such that $\text{p-tp}(\bar{a}) \setminus \text{p-tp}(f\bar{a}) \neq \emptyset$. To contradict 3, it suffices to extend f to a p-morphism defined on A . This is possible by ∞ -p-homogeneity.

$4 \Rightarrow 1$. Clear. □

10 Example Assume that $a \in \mathcal{N}$ is such that $\varphi(a; M^z)$ is definable over $\text{acl}^{\text{eq}} \emptyset$ for every $\varphi(x; z) \in L$. Then the orbit of a under $\text{Aut}(\mathcal{U}/\{M\})$ is p-maximal.

Proof. First consider the case when $\varphi(a; M^z)$ is definable over $\emptyset$. We prove that $\{a\}$ is p-maximal.

Let $\varphi'(z) \in L$ be such that $\varphi(a; M^z) \leftrightarrow \varphi'(M^z)$. Then, if we set $\varphi''(x; z) = \varphi(x; z) \leftrightarrow \varphi'(z)$, the formula $r_{\varphi''}(a)$ holds.

If f is a p-morphism then $r_{\varphi''}(fa)$ also holds. Then $a \equiv_{\Delta, M} b$ follows. This shows that p-morphisms are trivial and therefore p-isomorphisms.

Now assume that $\varphi(a; M^z)$ is definable over $\text{acl}^{\text{eq}} \emptyset$. □

A set $\mathcal{J} \subseteq \mathcal{U}^x$ is called a **core** if it is p-maximal and, among the p-maximal sets, it is maximal w.r.t. inclusion.

11 Fact Let $\mathcal{J}$ be a core. Let $\mathcal{A}$ be a p-maximal set. Then every p-morphism $f : \mathcal{A} \rightarrow \mathcal{J}$ extends to a p-isomorphism between $\mathcal{A}$ and $\mathcal{J}$.

Proof. Reason as in the proof of Fact 5 with the same notation but choose $b \in \mathcal{A}$. Let f' be a p-morphism defined on $\mathcal{A}$ that extends $f \cup \{\langle b, c \rangle\}$. Then $f'[\mathcal{A}]$, i.e. the image of $\mathcal{A}$ under f' , is a p-maximal set that contains $\mathcal{J}$. By maximality $f'[\mathcal{A}]$ coincides with $\mathcal{J}$, hence $c \in \mathcal{J}$. □

By back-and-forth we obtain the following.

12 Corollary (∞ -p-homogeneity of the core) If $\mathcal{J}$ and $\mathcal{J}'$ are cores, then every p-morphism $f : \mathcal{J} \rightarrow \mathcal{J}'$ extends to a p-isomorphism between $\mathcal{J}$ and $\mathcal{J}'$.

A **p-retraction** of $\mathcal{N}$ onto $\mathcal{A}$ is a total p-morphism $f : \mathcal{N} \rightarrow \mathcal{A}$ that is the identity on $\mathcal{A}$.

13 Theorem If $\mathcal{J}$ and is a core then there is a p-retraction of $\mathcal{N}$ onto $\mathcal{J}$

Proof. Let $\bar{c}$ enumerate $\mathcal{N}$. Let $\bar{a}$ be a tuple such that $\text{p-tp}(\bar{a}) \supseteq \text{p-tp}(\bar{c})$ is p-maximal. By Fact ?? the relation $R = (\equiv_M \circ \{\langle \bar{c}, \bar{a} \rangle\} \circ \equiv_M)$ is a p-morphism onto the p-maximal set $\mathcal{A} = \text{range } R$. As $\mathcal{A}$ is p-maximal, by Fact 11 we can extend $(R \cap (\mathcal{J} \times \mathcal{A}))^{-1}$ to a p-isomorphism R' defined on $\mathcal{A}$. It is easy to verify that $R' \circ R$ is a retraction. $\square$

14 Fact The following are equivalent

1. $a \in \bigcap \{\mathcal{J} : \mathcal{J} \text{ is a core}\}$
2. $\varphi(a; M^z)$ is definable over $\text{acl}^{\text{eq}} \emptyset$, for every formula $\varphi(x; z) \in L$.

Proof. ??? $\square$

15 Fact $\mathcal{J}$ is compact subset of $\mathcal{N}$.

Proof. ??? $\square$

6. Esperimenti

Let M be a saturated model and let $G = \text{Aut}(M)$.

The group G acts on $\mathcal{U}^x / \equiv_M$ in the natural way.